Geography of Colombian Life Satisfaction

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Happiness, Colombia, Encuesta Nacional de Calidad de Vida

One of the puzzles in the happiness literature is Latin American phenomenon (Rojas 2015, 2019): high happiness[1] despite low development in Latin America. Colombia is one of the happiest countries in the world–see table 1. But we know little about geography of happiness within Colombia. This paper is about geography of Colombian happiness across its geographies: departments, municipalities, and farmers v non-farmers.

Table 1: 10 happiest countries in the world. Data from World Database of Happiness (WDH) 2010-2019 out of 160 countries at `worlddatabaseofhappiness.eur.nl/rank-reports/satisfaction-with-life`; and World Values Surveys (WVS) 2005-2022 [waves 5-7] out of 88 countries at `https://www.worldvaluessurvey.org`.

WDH			WVS		
rank	country	happiness (1-10)	rank	country	happiness (1-10)
1	Denmark	8.2	1	Puerto Rico	8.4
2	Mexico	8.1	2	Mexico	8.3
3	Colombia	8.1	3	Colombia	8.3
4	Switzerland	8	4	Qatar	8.0
5	Finland	8	5	Norway	7.9
6	Iceland	8	6	Nicaragua	7.9
7	Costa Rica	7.9	7	Tajikistan	7.9
8	Norway	7.9	8	Switzerland	7.9
9	Canada	7.9	9	Uzbekistan	7.9
10	Qatar	7.8	10	Ecuador	7.8

A Peruvian social psychologist specializing in happiness argues that the origins of Latin happiness can be traced to:

> the minimalist well-being lessons of Andean and Amazonian small traditional communities which constitute the grounds of Latin American happiness, a life style that mimics the ancestral environment, the deep nature where the happiness brain wiring occurred; a physical and social environment that naturally activates the brain pleasure circuits. Culture resemble evolutionary needs; resources to achieve needs are available for everyone; positive, interdependent collectivistic interaction is ingrained in behavior, supporting, working, competing, and sharing. (Yamamoto 2016, p.45)

There is a Latin American concept of buen vivir (good living) that is about living close to nature and other people, in balance and sustainably, rejecting extractive materialistic and consumerist capitalism (Hidalgo-Capitán and Cubillo-Guevara 2017, Calisto Friant and Langmore 2015, Guardiola and García-Quero 2014, García-Quero and Guardiola 2018).

Nature and community contribute to SWB not just employment, income and consumption (Guardiola and García-Quero 2014). households that grow their own food and are in an indigenous community depend less on money to be happy (García-Quero and

[1]Following usual practice in social indicators research terms happiness and life satisfaction are used interchangeably, and specifically we mean cognitive and evaluative life satisfaction rather than affective happiness.

Guardiola 2018). Then we would expect that high happiness in Colombia will be found not only in richest parts of the country. Specifically, we would expect natural places (not large cities) to be happy. There could be some urban buen vivir especially iof city is not too large or dense and with good access to nature, and there could be lack of buen vivir in rural areas with industrial extractive agriculture and other problems. But in general, the more rural, the more nature and more buen vivir.

1 Literature

We know very little about geography of Colombian happiness–there are only 4 studies on the topic as far as we can tell: Peiró-Palomino et al. (2020)/Peiro-Palomino et al. (2021), PNUD (2023), Burger et al. (2021), and Okulicz-Kozaryn et al. (2026).

Peiró-Palomino et al. (2020)/Peiro-Palomino et al. (2021)[2] is a pioneering study here, but it uses an index objective indicators of quality of life such as income, housing and pollution. There is no subjective wellbeing studied here. Still, their study has a useful information for our study, especially the map in their Figure 1: Global well-being in the Colombian departments, 2016. Their objective wellbeing findings are quite similar to our subjective wellbeing findings.

Burger et al. (2021) does analyze subjective wellbeing geography within Colombia, but has few survey observations: only several thousand (7-9k depending on model), and problematic measurement across space–it uses questionable Gallup data (see discussion in Okulicz-Kozaryn and Valente 2021), hence, we leave it out. Okulicz-Kozaryn et al. (2026) only focuses or urban-rural. While it also breaks down the analyses by municipalities, it only addresses Bogota, and lumps all the other municipalities into 2 categories.

There is a recent United Nations study on Colombian happiness using the same Encuesta Nacional de Calidad de Vida as we use, just earlier waves (PNUD 2023). The study is available from the the UN.[3] It is closest to our study and most useful overview of Colombian happiness geography to date. There is in fact a similar map to ours in Figure 1.9 on page 26 for 2018-2021.While happy Cafetero agrees with our analysis, happy Narino does not. Likewise it finds Guaviare to be unhappy, while we find it moderately happy. A possible explanation for the differences is different time-frame: 2018-2021 theirs v 2023 and 2021 ours. Likewise in figure 1.10 they break down SWB by urban-rural, our results agree that there is almost no difference, and in addition to urban-rural breakdown we add farmer-nonfarmer breakdown and add income breakdown for both. The study also does analysis for municipalities as we do in figures 2.14, 2.17 and 2.21 but the analyses do not address SWB. And only cover 10 cities, while we cover 50+.

Overall, while there are at least 4 studies on the topic, only one (PNUD 2023) really addresses the life satisfaction geography of Colombia, and we expand and improve on this study.

2 Data

We use a great source of data Encuesta Nacional de Calidad de Vida (ECV). We use 2023 edition, and in appendix for robustness check 2021. Sampling details are at `https://microdatos.dane.gov.co/index.php/catalog/734`

The ECV is nationally representative, cross-sectional annual survey in Colombia, it includes about 250k persons and 80k households across all 32 Colombian departments and rural and urban.[4]

The ECV identifies municipalities, but it is representative of departments, not municipalities,[5] hence municipalities results are provisional and should be interpreted with care. Still, the samples are sizable and we are able to use many municipalities with at least few hundred observations.

[2] Both are the same study.

[3] `https://www.undp.org/sites/g/files/zskgke326/files/2023-02/undp_co_pub_Percepciones_bienestar_subjetivo_2do_cuaderno_INDH_Feb21_2023.pdf`.

[4] `https://www.dane.gov.co/index.php/estadisticas-por-tema/salud/calidad-de-vida-ecv/encuesta-nacional-de-calidad-de-vida-ecv-2021`

[5] See `https://microdatos.dane.gov.co/index.php/catalog/905` and `https://www.dane.gov.co/index.php/estadisticas-por-tema/salud/calidad-de-vida-ecv/encuesta-nacional-de-calidad-de-vida-ecv-2024`.

Income variable comes from module 'Servicios del hogar,' and all other variables come from module 'Caracteristicas y composicion del hogar.'

`'Life satisfaction'` is measured with P1895 'Overall, how satisfied do you feel ... with your life at present?' on scale from 0='Completely dissatisfied' to 10='Completely satisfied.' The actual question asked in Spanish was: 'En general, qué tan satisfecho(a) se siente ... con su vida actualmente? (P1895)' on scale from 0='Totalmente insatisfecho(a)' to 10= 'Totalmente satisfecho(a)' `'Farmer'` is measured with p2057 'Do you consider yourself a farmer?' (¿Usted se considera campesino(a)? 1=Si 0=No) `'Rural'` is measured with P2061 'Do you consider the community where you live to be a rural farming community?' (¿Usted considera que la comunidad en que vive es campesina? 1=Si 0=No) `'Income'` is 'per capita income' (Ingreso per-capita).

One problem with ECV is missing values. There are 166k missing out of 242k on the variables of interest. This is one reason we conduct robustness check in appendix using 2021 wave.

3 Departments/Provinces

We map `life satisfaction` by department in figure 1. The table including number of observations per province is in appendix appendix. Here we present results for 2023, appendix has results for 2021. Finally, appendix section 'Administrative Boundaries' shows maps with department names.

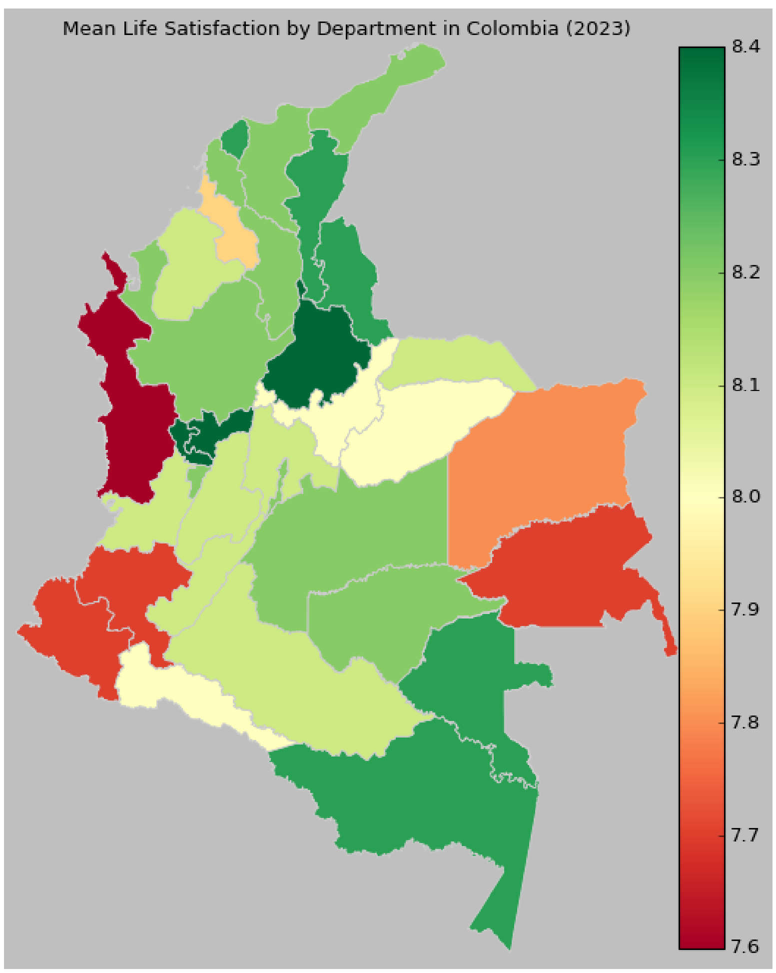


Figure 1: Life satisfaction across departments

There are quite large differences close to 1 on 0-10 life satisfaction scale. Unhappy departments are: West (poor Choco, and

Southern Cauca and Narino) and East Guainia and Vichada. In 2021 (see appendix) Choco and Cauca were still relatively unhappy, but slightly happier.

Happy departments are in Cafetero: Caldas, Risaralda, Quindio–in 2021 there were slight differences (see appendix) but overall Cafetero was happy too. And also Santander is happy. Then slightly less happy Norte de Santander, Cesar and Atlantico. In 2021 (see appendix) Santander, Cesar and Atlantico were less happy.

Caraibean Cordoba and Sucre are relatively unhappy, but then Bolivar where Caratgena is located is quite happy and so Atlantico where is Barranquilla, at least in 2023 (not so much in 2021).

Overall, figure 1 2023 results are similar to 2021 robustness/sensitivity check in appendix except: small sample Amazonas and Vaupés, both South-Eastern at 8.3 in 2023 v 7.6 in 2021, a .7 difference. Second largest differences: Santander and Putumayo increased by .4 from 2021 to 2023.

4 Municipalities

Next we turn to municipalities in figure 2. Appendix includes table with municipalities names and life satisfaction values. Here the differences even larger[6] than across provinces at over 1 on 0-10 life satisfaction scale.

[6] As expected, smaller geographies have more variability.

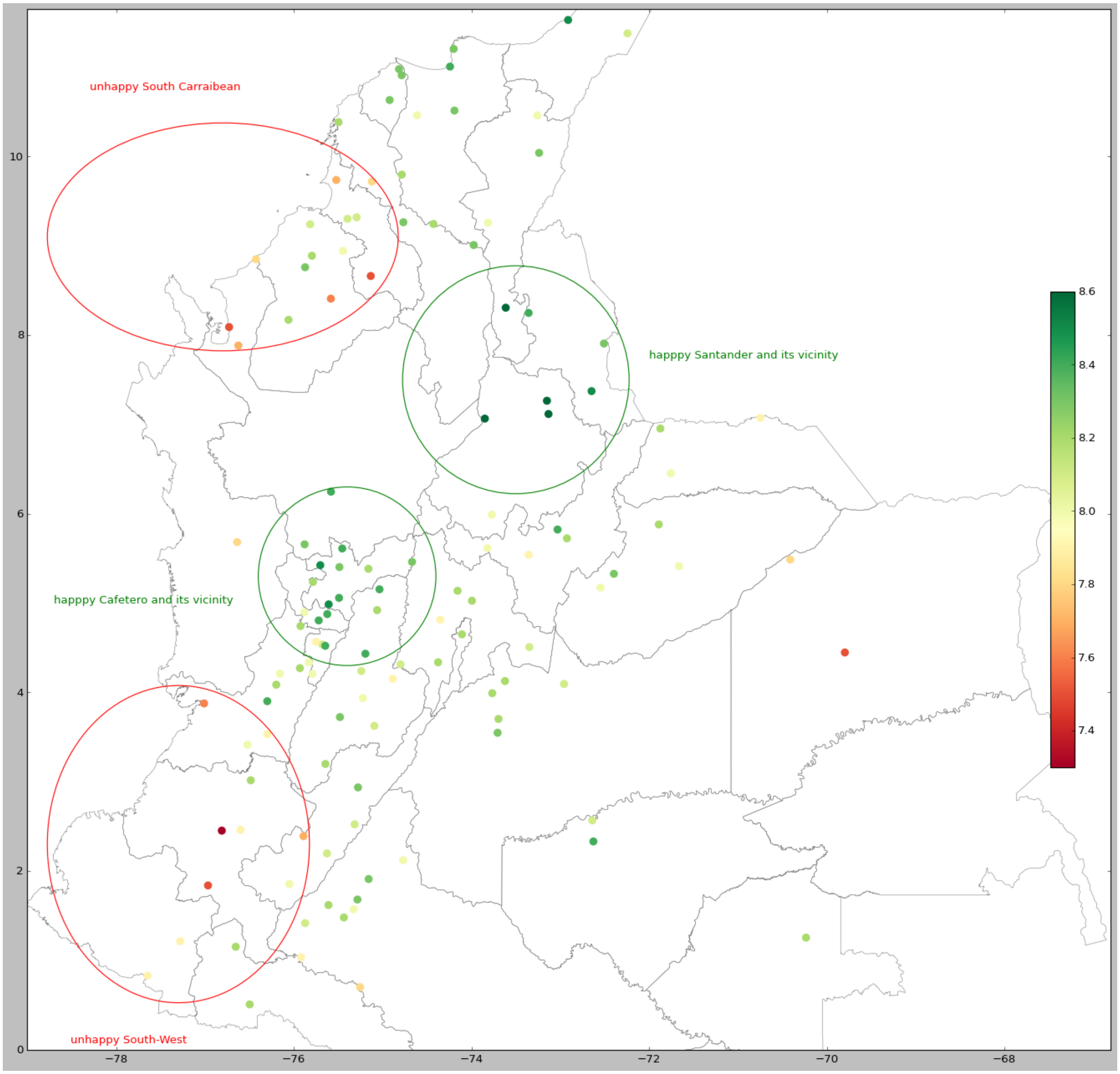


Figure 2: Life satisfaction across municipalities. Zoom in to see the detail.

Santander is one of the happiest departments, and so 2 Santander largest cities shine: Bucaramanga and Barrancabermeja at 8.6. Bucaramanga was less happy in 2021 at 8.1. Another happy region Cafetero has many happy cities: Calarcá, Pereira, Manizales, and Armenia (Armenia was even happier in 2021). Other happy cities include 2nd largest Medellin, Ibague, and Ciénaga (Ciénaga was less happy in 2021 at 7.9). The least happy is remote Western Puerto Carreño at 7.5, and several cities at 7.9: Tunja, and Southern cities: Palmira in Valle de Cauca, Popayan in Cauca, and Pasto in Narino.

In 2021 appendix results are similar, except that Santander and its vicinity is less happy.

5 Farmers v Non-farmers

While most developed countries cities are unhappy and indeed unhappy city is a common pattern across the world (Okulicz-Kozaryn and Valente 2021), it is not always the case in developing countries.

Cities are economic growth engines (Glaeser 2011, O'Sullivan 2009) and developing countries like Colombia do need economic growth hence role for cities is greater than in already developed countries. And there is a "paradox"[7] of happy peasant and unhappy millionaire Graham (2009). In Colombia, we do not find much evidence for the paradox of happy peasant and unhappy millionaire–urban and rural are about same happy, except by income categories.

We plot life satisfaction for farmers v non-farmers in figures 3 and 4–there are slight differences, but overall similar pattern in both figures. There is no difference for the bottom and 9th deciles; rural and farmer happier in 2nd-8th deciles, and non-farmer happier in the top decile. This makes sense, one needs money to enjoy city, where cost of living is higher. In addition, life is more commodified in the city–one needs to pay for more things. For instance, many in the rural areas grow their own food (García-Quero and Guardiola 2018). The bottom decile parity could be explained as that there is some minimum income necessary, and below it, it doesn't help to be a farmer v non-farmer.

In other words, there are lower returns in happiness from income for rural (farmers) v urban persons (non-farmers), or there is a bigger premium from income in urban.

One explanation could be that money can buy more varied and luxury goods and services in urban than rural, and one can enjoy more material comfort and consumption. Urbanity also provides more anonymity and freedom (Okulicz-Kozaryn 2015) and perhaps in such conditions more money can translate into more happiness. Another explanation is that non-farm and urban areas cost of living is higher. Likewise, in rural areas many goods and services are outside of the official economy. For instance, many grow their own food, have hens that lay eggs, use freely available materials such as wood, stones, and sand for construction, and so forth. Households that grow their own food and are in an indigenous community depend less on money to be happy (García-Quero and Guardiola 2018)

Overall, it is worth emphasizing that except for the 1st decile and up to median and one decile above it–farmers/rural clearly happier.

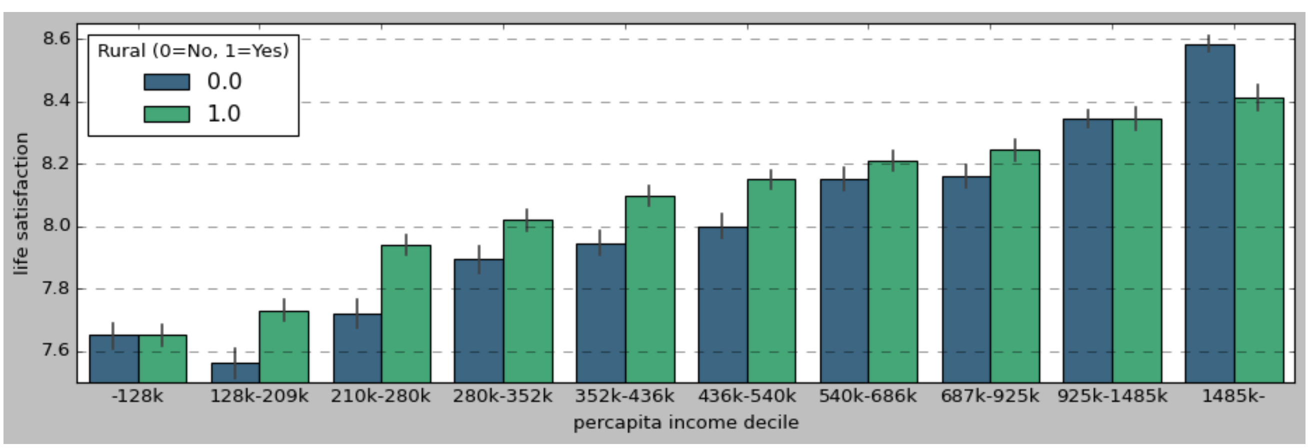


Figure 3

[7]Not so much paradox if you think about it there is actually little reason that a millionaire should be happier than a peasant (Okulicz-Kozaryn and Martinez 2025).

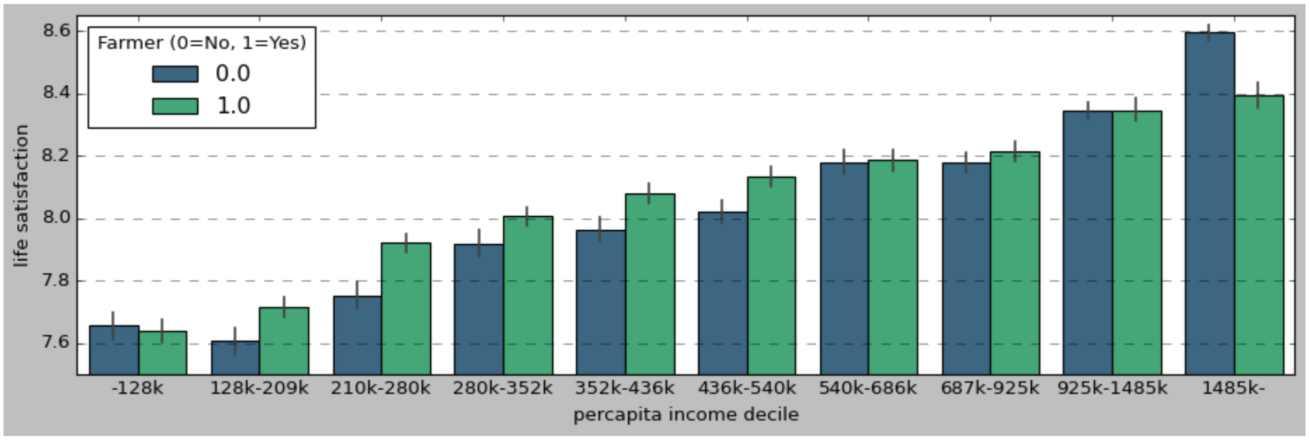


Figure 4

For rural in 2021 (appendix) pattern is similar but less strong for lower deciles where rural are happier than urban, but stronger for 2 top deciles where the opposite is true. For farmers in 2021 there is no happiness premium in lower deciles, but for non-farmers there is happiness premium in 2 top deciles.

6 Conclusion and Discussion

Colombia, one of the very happiest countries in the world, has large happiness differences across geography within the country. Differences across the 32 departments are as large as close to 1 on 0-10 SWB scale. Differences across municipalities that were covered in the data are even larger at above 1 on 0-10 SWB scale. Farmers and non-urban settlements are happier than non-farmers and urbanity across income distribution (supporting Buen Vivir hypothesis) except first and last deciles. In last decile non-farmers and city dwellers are significantly happier–perhaps high income goes a longer way for non-farmers and cities. Another explanation is that non-farm and urban areas cost of living is higher.

We have used a fantastic and underused dataset, Encuesta Nacional de Calidad de Vida (ECV)–it has tens of thousands of observations and hundreds of variables. Colombia is the leader in SWB data collection.

This is the first study focusing on geography of Colombian happiness–the purpose here was to map the data and present basic relationships in space. Future research can look by socio-demographic subgroups, say by age, education, and income–how different groups score on SWB across space. It is likely that younger, more educated and richer folks are happier in cities v rural areas. We have already found some evidence here with respect to income–for the richest income decile there was a happiness premium for non-farmers.

ONLINE APPENDIX (Supplementary Online Material (SOM)

[note: this section will NOT be a part of the final version of the manuscript, but will be available online instead]

7 Additional Ecological Predictors at Department Level

7.1 PCGDP

There is a positive and quadratic relationship between PCGDP and SWB, but there is a high spread in SWB along quadratic fit in figure 5. Among departments with low PCGDP of 2-5k usd, there is a large variability in SWB: ranging from least happy Choco at 7.6

to quite happy Vaupes at 8.3. And then at the high-end of PCGDP of 13-15, there is moderately happy Casanare at 8 to very happy Santander at 8.4.

There is also large spread horizontally: many departments with the same SWB on y-axis have very different PCGDP levels on x-axis.

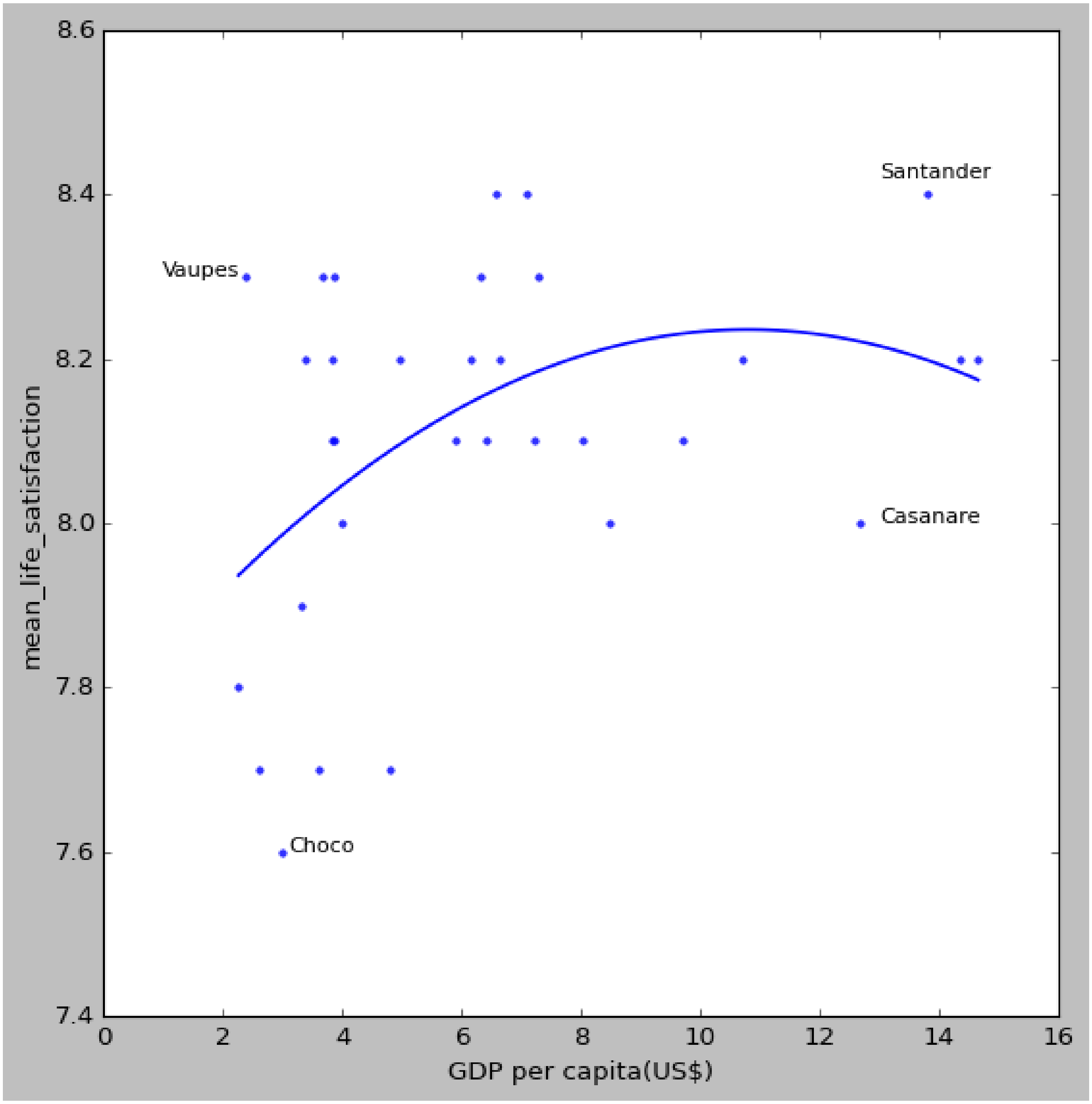


Figure 5

7.2 Population Density

excluded Bogota because it is not a regular department but capital and San Andres, which is an island. On populatiin density in figure 6 a clear outlier is Atlantico with density at 750, more than double 2nd Quindio at about 300.

Most departments are very sparsely populated at below or around 100, and there is a great variability in SWB among them from unhappy Choco at ?? to ?? at ??. The denser departments at around 200-300 tend to quite happy at >8.1, but some less dense departments are also at >8.1.

The positive relationship would contradict happy farmer hypothesis, however, departments tend ot be large and there is a wide variety of density within, hence the happy farmer hypthesis is not necessarily contradicted

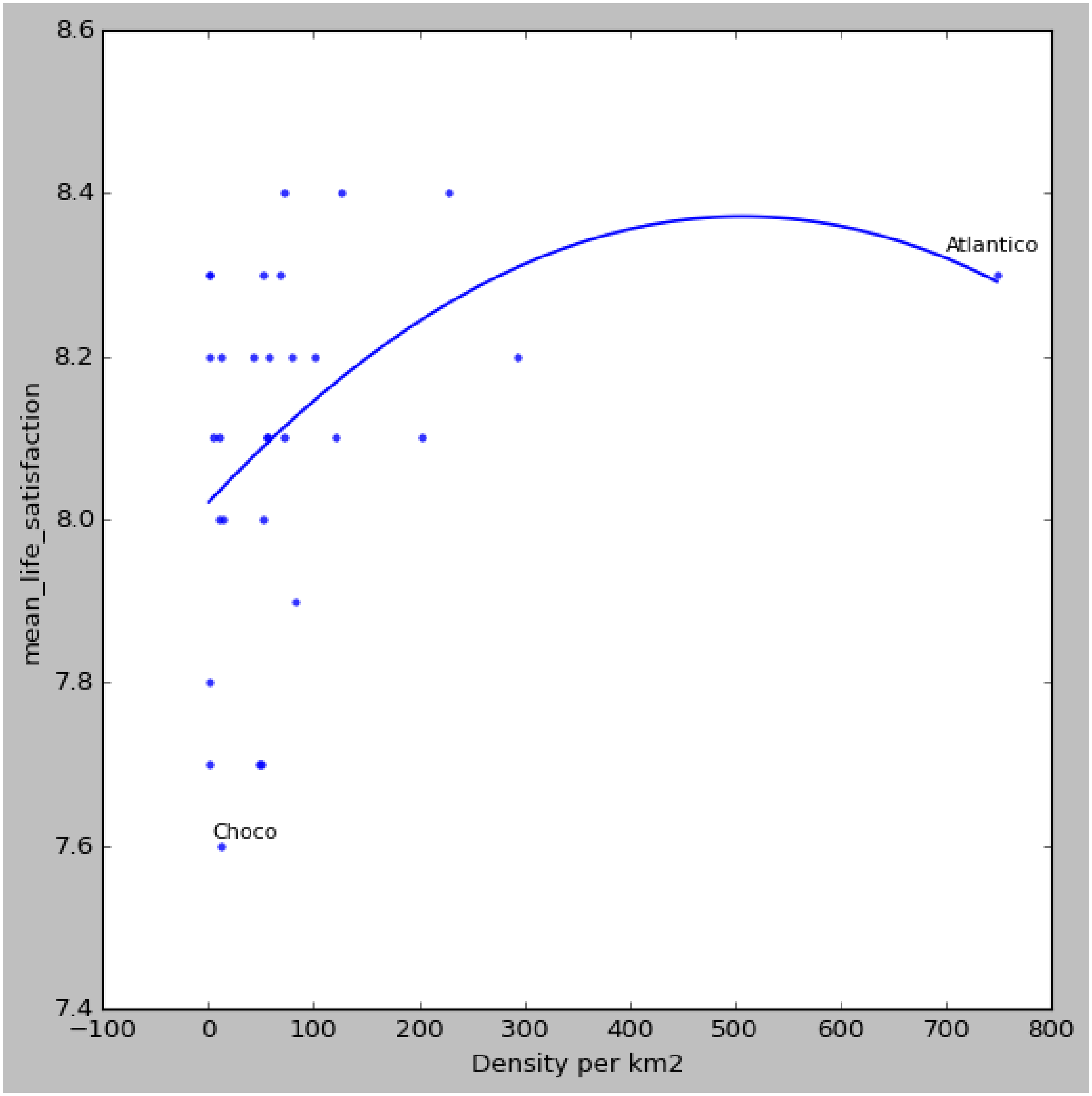


Figure 6

7.3 Crime

the relationship is inverted u-shape in figure 7, opposed to expected negative. However, again, departments tend ot be large and there is a wide variety of crime within.

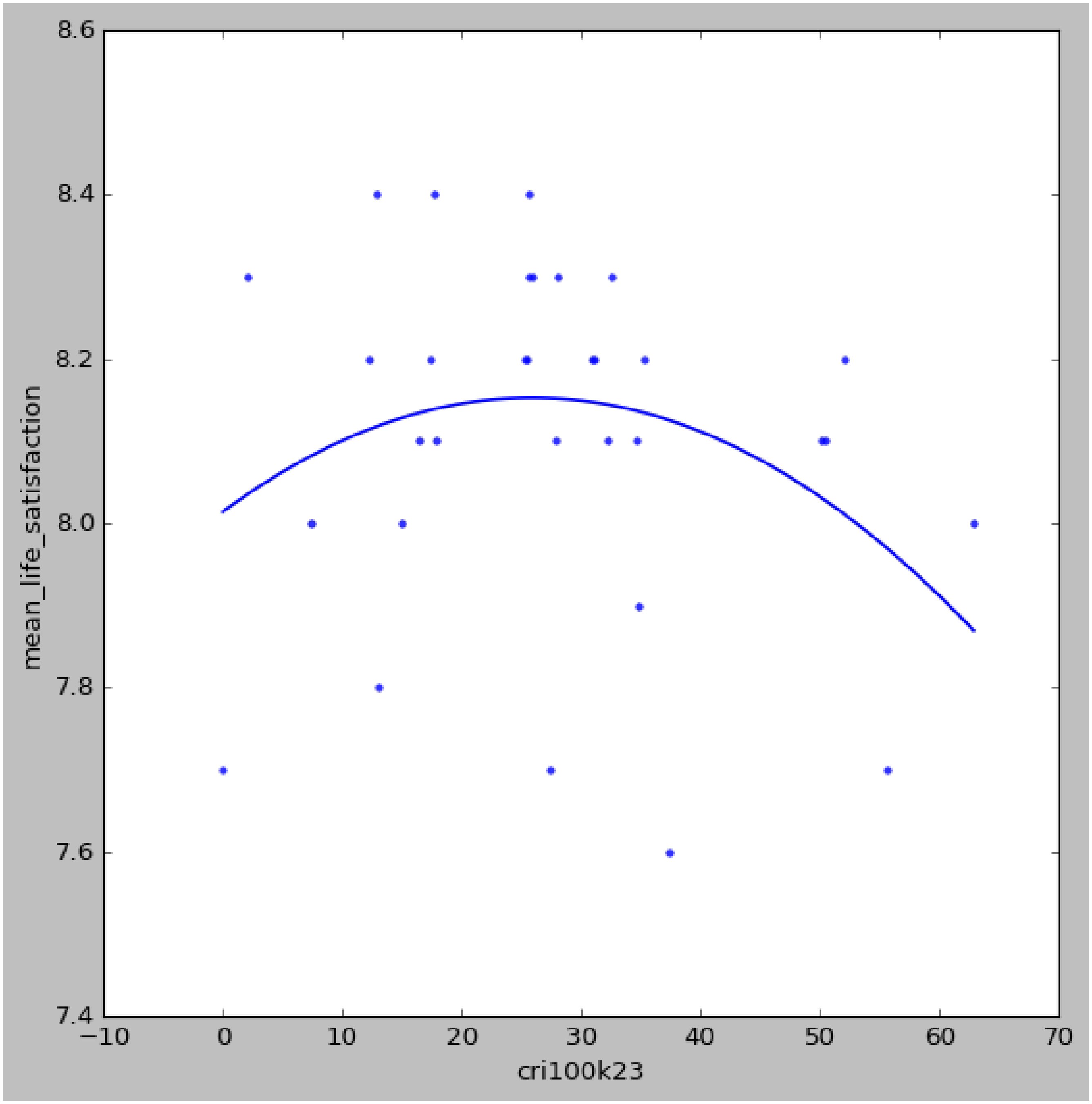


Figure 7

8 Re-estimation Using 2021 ECV

2021 May be biased say because of covid, especially that it peaked in Colombia in 2021 `https://coronavirus.jhu.edu/region/colombia`(by 2023 used in the paper it pretty much cleared). But it a useful robustness check, if covid didn't change much , then it should be pretty stable

and it was quite high mortality in col `https://coronavirus.jhu.edu/data/mortality`

8.1 Departments/Provinces

Again as said in the paper, overall results are similar to 2023 reported in the body of the paper, except both small sample Amazonas and Vaupés, South-Eastern and with 8.3 in 2023 v 7.6 in 2021, a .7 difference. Second largest differences are for Santander and Putumayo which increased by .4 from 2021 to 2023.

Table 2: Life satisfaction by department.

department	life satisfaction 2023	total obs 2023	non-missing life satisfaction 2023	mean life satisfaction 2021	total obs 2021	non missing life satisfaction 2021	2023-2021
Amazonas	8.3	276	245	7.6	423	365	0.7
Antioquia	8.2	5499	4729	8.3	6036	5060	-0.1
Arauca	8.1	974	773	8.1	1103	868	0
Atlántico	8.3	3144	2311	8.0	2338	1799	0.3
Bogotá, D.C.	8.2	4688	3446	8.1	5337	3810	0.1
Bolívar	8.2	2584	2258	8.1	2495	2156	0.1
Boyacá	8.0	4129	3649	7.8	4638	4122	0.2
Caldas	8.4	2822	2555	8.2	3335	3005	0.2
Caquetá	8.1	2547	2018	7.9	2846	2249	0.2
Casanare	8.0	1749	1370	8.1	2051	1457	-0.1
Cauca	7.7	2580	2244	7.9	2572	2287	-0.2
Cesar	8.3	2224	1717	8.0	1833	1510	0.3
Chocó	7.6	857	754	7.9	1016	913	-0.3
Cundinamarca	8.1	3818	3322	7.8	4233	3684	0.3
Córdoba	8.1	2706	2240	8.0	2226	1933	0.1
Guainía	7.7	327	273	7.4	350	274	0.3
Guaviare	8.2	700	457	8.0	621	411	0.2
Huila	8.1	2961	2322	8.0	2999	2407	0.1
La Guajira	8.2	951	750	7.9	988	736	0.3
Magdalena	8.2	2458	2096	8.0	2110	1710	0.2
Meta	8.2	3683	2752	8.1	3862	2817	0.1
Nariño	7.7	2284	2034	7.4	2847	2492	0.3
Norte de Santander	8.3	2499	2028	8.2	2587	2171	0.1
Putumayo	8.0	950	752	7.6	1272	1012	0.4
Quindío	8.2	2085	1658	8.4	2186	1785	-0.2
Risaralda	8.4	1634	1353	8.3	1996	1663	0.1
San Andrés y Providencia	8.4	28	25	8.2	29	26	0.2
Santander	8.4	3790	3194	8.0	4153	3574	0.4
Sucre	7.9	1798	1399	7.8	1899	1467	0.1
Tolima	8.1	3564	3215	8.0	4257	3770	0.1
Valle del Cauca	8.1	4892	4093	8.1	5268	4421	0
Vaupés	8.3	265	245	7.6	288	267	0.7
Vichada	7.8	515	455	7.6	584	482	0.2

8.2 Cities/Municipalities

Very important this robustness check because ECV is not representative of municipalities.

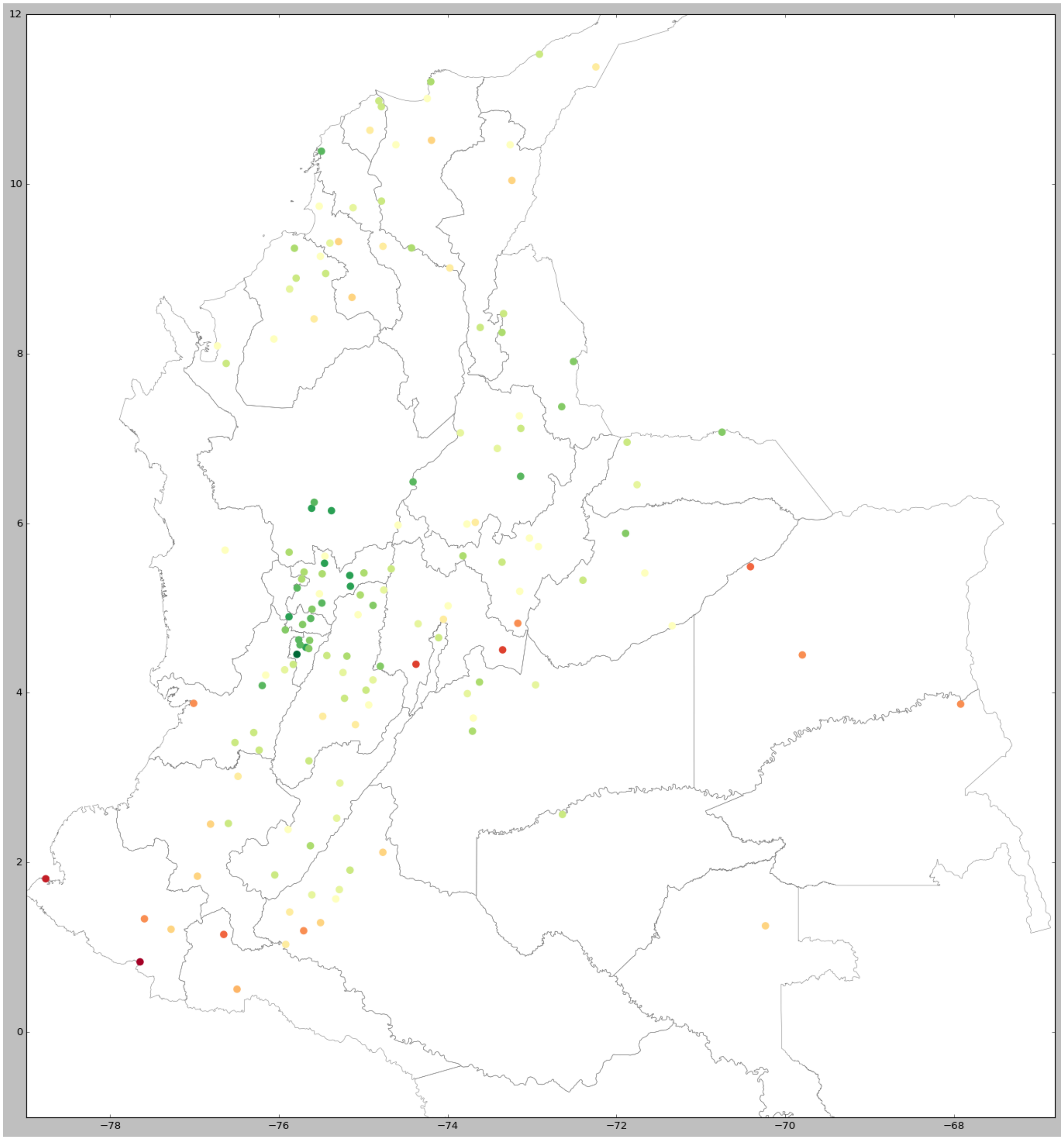


Figure 8: Zoom in to see the detail

Table 3: Life satisfaction by municipality, where non-missing on life satisfaction >200 or 220??

municipality name	mean life satisfaction 2023	non missing life satisfaction 2023	mean life satisfaction 2021	non missing life satisfaction 2021	2023-2021 diff
Buga	NaN	NaN	8.0	215	NaN
Espinal	NaN	NaN	7.8	220	NaN
Mariquita	NaN	NaN	7.9	201	NaN
Ocaña	NaN	NaN	8.2	200	NaN
San Vicente del Caguán	NaN	NaN	8.1	240	NaN
Armenia	8.1	668.0	8.5	681	-0.4
Buenaventura	8.1	412.0	8.3	279	-0.2
Cartagena	8.2	591.0	8.4	572	-0.2
Facatativá	8.1	200.0	8.3	222	-0.2
Palmira	7.9	319.0	8.1	332	-0.2
Popayán	7.9	225.0	8.1	232	-0.2
Tuluá	8.2	303.0	8.4	307	-0.2
Tunja	7.9	270.0	8.1	241	-0.2
Cali	8.0	985.0	8.1	1108	-0.1
Pitalito	8.0	293.0	8.1	288	-0.1
Cúcuta	8.3	573.0	8.3	553	0.0
Manizales	8.4	462.0	8.4	567	0.0
Medellín	8.4	905.0	8.4	959	0.0
Puerto Carreño	7.5	225.0	7.5	215	0.0
San José del Guaviare	8.1	246.0	8.1	231	0.0
Villavicencio	8.2	982.0	8.2	990	0.0
Acacías	8.3	229.0	8.2	233	0.1
Bogotá, D.C.	8.2	3446.0	8.1	3836	0.1
Calarcá	8.4	244.0	8.3	257	0.1
Pereira	8.4	524.0	8.3	677	0.1
Santa Marta	8.3	225.0	8.2	218	0.1
Sincelejo	8.1	222.0	8.0	275	0.1
Valledupar	8.0	339.0	7.9	274	0.1
Barranquilla	8.3	1363.0	8.1	1116	0.2
Florencia	8.2	575.0	8.0	630	0.2
Ibagué	8.4	478.0	8.2	549	0.2
Pasto	7.9	244.0	7.7	309	0.2
Yopal	8.3	264.0	8.1	240	0.2
Montería	8.3	465.0	8.0	372	0.3
Neiva	8.3	417.0	8.0	436	0.3
Riosucio	8.5	203.0	8.2	203	0.3
Sogamoso	8.2	236.0	7.9	367	0.3
Aguachica	8.6	202.0	8.1	215	0.5
Bucaramanga	8.6	798.0	8.1	842	0.5
Ciénaga	8.4	234.0	7.9	252	0.5
Magangué	8.3	205.0	7.8	207	0.5

and here below ¿100

Table 4: Life satisfaction by municipality, where non-missing on life satisfaction ¿200 or 220??

municipality name	mean life satisfaction 2023	non missing life satisfaction 2023	mean life satisfaction 2021	non missing life satisfaction 2021	2023-2021 diff
Acacías	8.3	229.0	8.2	233.0	0.1
Aguachica	8.6	202.0	8.1	215.0	0.5
Aguadas	8.4	153.0	7.9	161.0	0.5
Aguazul	8.2	143.0	8.3	171.0	-0.1
Aguazul	8.0	102.0	8.3	171.0	-0.3
Aipe	8.1	112.0	8.0	100.0	0.1
Anserma	8.2	129.0	8.4	174.0	-0.2
Apartadó	8.3	119.0	8.2	111.0	0.1
Apartadó (Chigorodó)	7.8	132.0	NaN	NaN	NaN
Aracataca	8.3	190.0	7.7	122.0	0.6
Arauca	7.9	199.0	8.3	199.0	-0.4
Armenia	8.1	668.0	8.5	681.0	-0.4
Ataco	NaN	NaN	8.3	101.0	NaN
Barichara	8.0	131.0	7.9	128.0	0.1
Barrancabermeja	8.6	270.0	8.0	199.0	0.6
Barranquilla	8.3	1363.0	8.1	1116.0	0.2
Bello	NaN	NaN	8.5	103.0	NaN
Bello (Copacabana)	NaN	NaN	8.5	103.0	NaN
Bogotá, D.C.	8.2	3446.0	8.1	3836.0	0.1
Bosconia	8.3	106.0	7.7	124.0	0.6
Bucaramanga	8.6	798.0	8.1	842.0	0.5
Buenaventura	8.2	220.0	8.1	166.0	0.1
Buenaventura	8.0	192.0	8.1	166.0	-0.1
Buenaventura	8.2	220.0	8.3	279.0	-0.1
Buenaventura	8.0	192.0	8.3	279.0	-0.3
Buga	8.2	193.0	8.0	215.0	0.2
Cajibío	7.5	152.0	7.7	167.0	-0.2
Cajicá	NaN	NaN	7.8	106.0	NaN
Calamar	8.4	105.0	NaN	NaN	NaN
Calarcá	8.4	244.0	8.3	257.0	0.1
Cali	8.0	985.0	8.1	1108.0	-0.1
Candelaria	8.0	113.0	7.9	154.0	0.1
Cartagena	8.2	591.0	8.4	572.0	-0.2
Cartagena del Chairá	8.1	120.0	7.8	136.0	0.3
Cartago	7.6	168.0	7.5	187.0	0.1
Cereté	8.2	131.0	8.1	109.0	0.1
Chaparral	NaN	NaN	7.9	108.0	NaN
Chinchiná	8.5	141.0	8.3	198.0	0.2
Chiquinquirá	8.4	128.0	7.9	152.0	0.5
Circasia	8.0	102.0	NaN	NaN	NaN
Ciénaga	8.4	234.0	7.9	252.0	0.5
Codazzi	8.0	103.0	NaN	NaN	NaN
Corozal	8.1	174.0	7.7	167.0	0.4
Coyaima	NaN	NaN	8.1	110.0	NaN
Curillo	7.9	103.0	7.8	123.0	0.1
Cúcuta	8.3	573.0	8.3	553.0	0.0
Cúcuta (El Zulia)	NaN	NaN	8.1	106.0	NaN
Dosquebradas	8.0	103.0	8.5	141.0	-0.5
Duitama	8.0	134.0	8.2	138.0	-0.2
El Carmen de Bolívar	7.8	200.0	8.0	167.0	-0.2
El Doncello	8.3	158.0	8.0	160.0	0.3
El Paujil	8.0	100.0	7.9	106.0	0.1
Envigado	7.7	146.0	8.1	180.0	-0.4
Espinal	8.3	170.0	7.8	220.0	0.5
Facatativá	8.1	200.0	8.3	222.0	-0.2
Flandes	7.9	135.0	8.0	195.0	-0.1
Florencia	8.2	575.0	8.0	630.0	0.2
Florida	8.4	119.0	NaN	NaN	NaN
Fundación	8.3	200.0	7.8	155.0	0.5
Fusagasugá	8.2	156.0	7.3	117.0	0.9
Garagoa	NaN	NaN	7.9	109.0	NaN
Garzón	8.1	181.0	8.2	161.0	-0.1
Girardot	8.1	133.0	7.3	151.0	0.8
Granada	8.2	186.0	8.0	185.0	0.2
Guachené	7.3	138.0	7.7	161.0	-0.4
Guamo	NaN	NaN	8.1	106.0	NaN
Génova	NaN	NaN	8.3	101.0	NaN
Honda	NaN	NaN	8.0	102.0	NaN
Ibagué	8.4	478.0	8.2	549.0	0.2
Inírida	7.8	102.0	7.4	120.0	0.4
Inírida	7.8	102.0	7.5	149.0	0.3
Ipiales	7.9	137.0	7.1	134.0	0.8
Itagüí	NaN	NaN	8.4	128.0	NaN
Jamundí	NaN	NaN	8.1	103.0	NaN
La Calera	8.2	105.0	NaN	NaN	NaN
La Dorada	8.3	189.0	8.1	197.0	0.2
La Montañita	8.2	106.0	NaN	NaN	NaN
La Plata	7.7	133.0	7.9	128.0	-0.2
La Tebaida	NaN	NaN	8.7	113.0	NaN
Lorica	8.1	201.0	8.2	150.0	-0.1
Líbano	8.4	116.0	8.2	113.0	0.2
Magangué	8.3	205.0	7.8	207.0	0.5
Maicao	8.1	167.0	7.8	162.0	0.3
Malambo	8.3	234.0	7.8	136.0	0.5
Manizales	8.4	462.0	8.4	567.0	0.0
Manzanares	NaN	NaN	7.9	103.0	NaN
Mariquita	8.2	173.0	7.9	201.0	0.3
Marmato	NaN	NaN	8.5	121.0	NaN
Marulanda	NaN	NaN	8.5	100.0	NaN
Medellín	8.4	905.0	8.4	959.0	0.0
Mitú	8.2	173.0	7.7	181.0	0.5
Mocoa	8.2	112.0	7.4	128.0	0.8
Mompós	8.2	124.0	8.2	135.0	0.0
Montenegro	7.9	146.0	8.4	176.0	-0.5
Montería	8.3	465.0	8.0	372.0	0.3
Morelia	NaN	NaN	7.7	111.0	NaN
Natagaima	8.1	103.0	7.8	102.0	0.3
Neira	8.2	109.0	8.5	114.0	-0.3
Neiva	8.3	417.0	8.0	436.0	0.3
Ocaña	8.4	182.0	8.2	200.0	0.2
Paipa	NaN	NaN	7.9	104.0	NaN
Palmira	7.9	319.0	8.1	332.0	-0.2
Pamplona	8.5	130.0	8.3	116.0	0.2
Pasto	7.9	244.0	7.7	309.0	0.2
Paz de Ariporo	NaN	NaN	7.9	113.0	NaN
Pereira	8.4	524.0	8.3	677.0	0.1

8.3 Urban-Rural and Farmers

Colombia is quite urbanized at around 70-80%. `https://population.un.org/wup/countryprofiles?country=Colombia&mainCategory` `Countries%20and%20Aggregates&subCategory=Urbanization%20-%20International%20definition`

Still, a substantial part at 20-30% is rural.

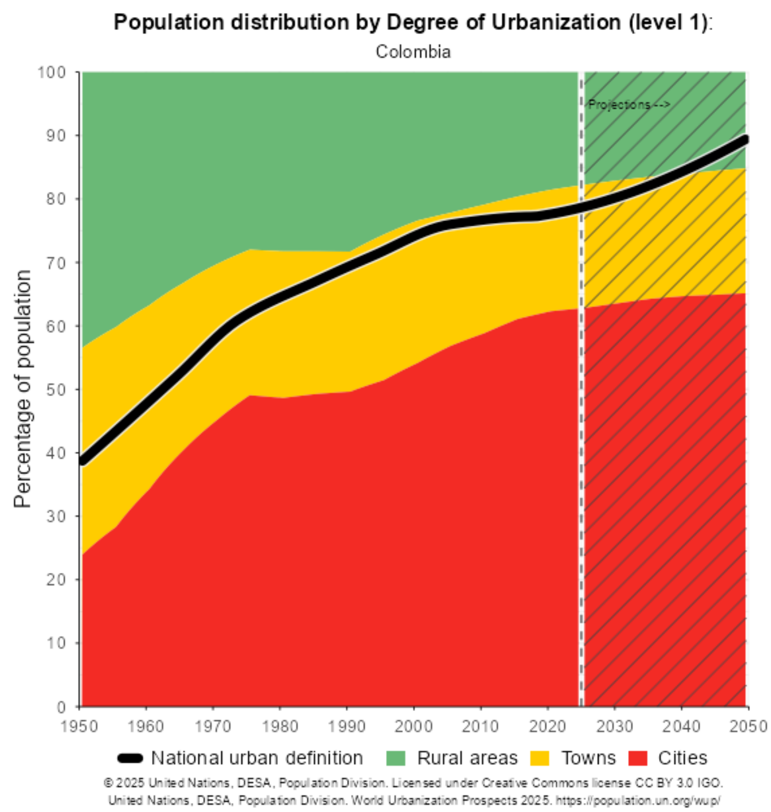


Figure 9

For rural in 2021 pattern is similar to 2023 in body of the paper but less strong for lower deciles where rural are happier than urban, but stronger for 2 top deciles where the opposite is true. For farmers in 2021 there is no happiness premium in lower deciles, but for non-farmers there is happiness premium in 2 top deciles.

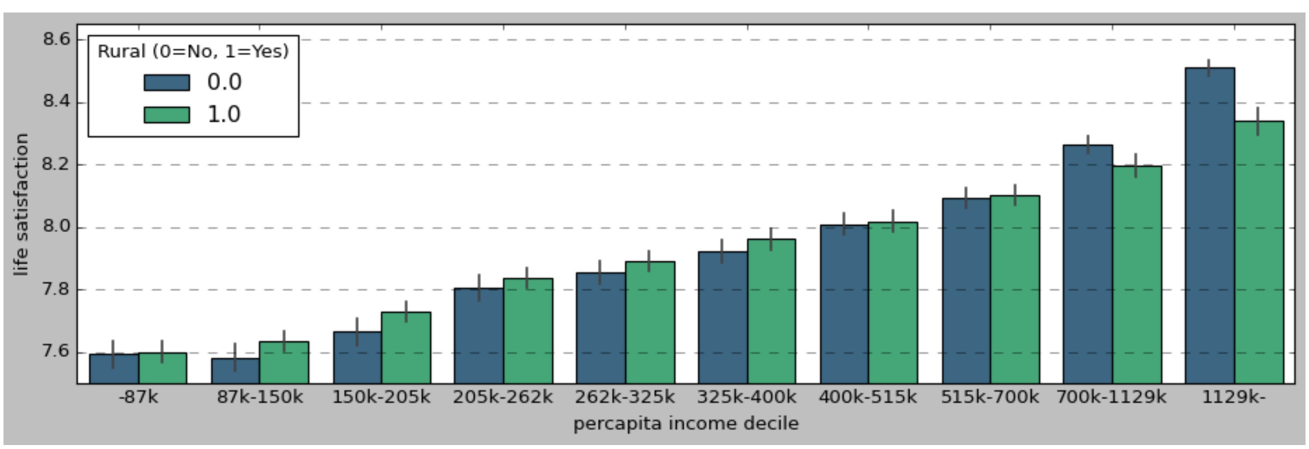


Figure 10

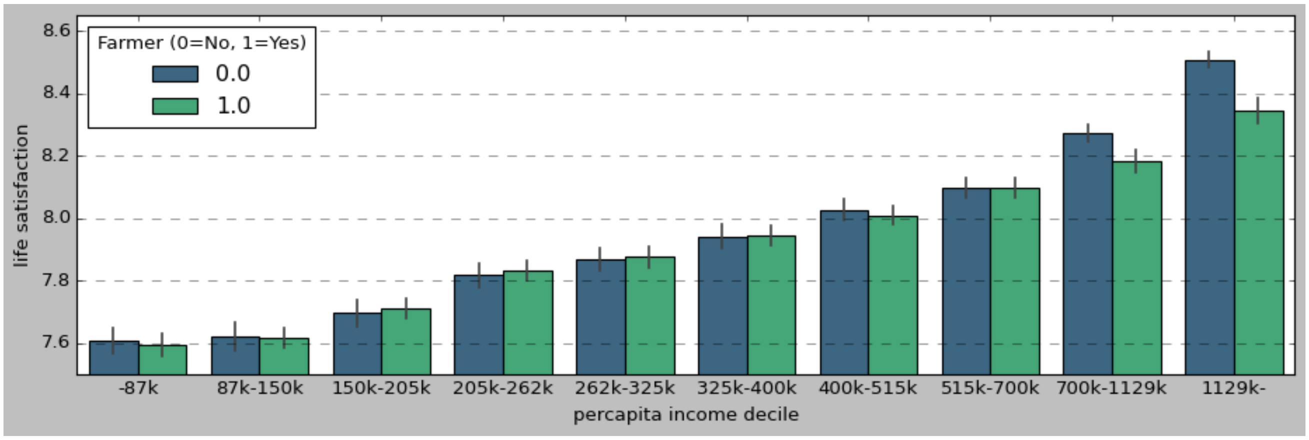


Figure 11

9 Geography of Colombia

9.1 Administrative Boundaries

Colombia is at North-West of South America (fig 12), a large country of about 54m, third largest in Latin America after Brazil and Mexico (`https://fred.stlouisfed.org/series/POPTOTCOA647NWDB`.) Colombia is also large in land mass at about 1.1m sq km, while much smaller than Brazil at 8.5m sq km, it is larger than Venezuela at .9m sq km, and about 3x larger than Germany at .35m sq km and about 2x of Spain at .5 sq km.

Figure 12: Colombia circled in red in Latin America shaded in green.

Map of the 32 departments is in figure 13.

Figure 13: Colombia 32 departments. Source: `https://upload.wikimedia.org/wikipedia/commons/thumb/6/67/Departments_of_colombia.svg/960px-Departments_of_colombia.svg.png`.

TODO repeat both 2 maps from the body with dept name and city names and still thematic on happiness!

9.2 Physical Geography

The paper is about happiness geography, but it would be incomplete without physical geography. Topography is well mapped and best viewed at `https://en.wikipedia.org/wiki/File:Mapa_de_Colombia_(relieve).svg`, where you can zoom in to see the detail. Population is nicely visualized in a 3d map at `https://geoportal.dane.gov.co/geovisores/sociedad/grilla-dane-vihope/`. To compare topography v population density here side by side see figure 14. Much of the population is concentrated around montaneous areas (with a notable exception of quite populated and largely flat Caraibean region in North).

Colombia's road infrastructure and connectivity is very poor.[8] Then this could be one explanation for large distinctivness/uniqness of regions–they have not become homogenous as in many other countries as they are weakly connected due to mountains and poor road infrastructure.

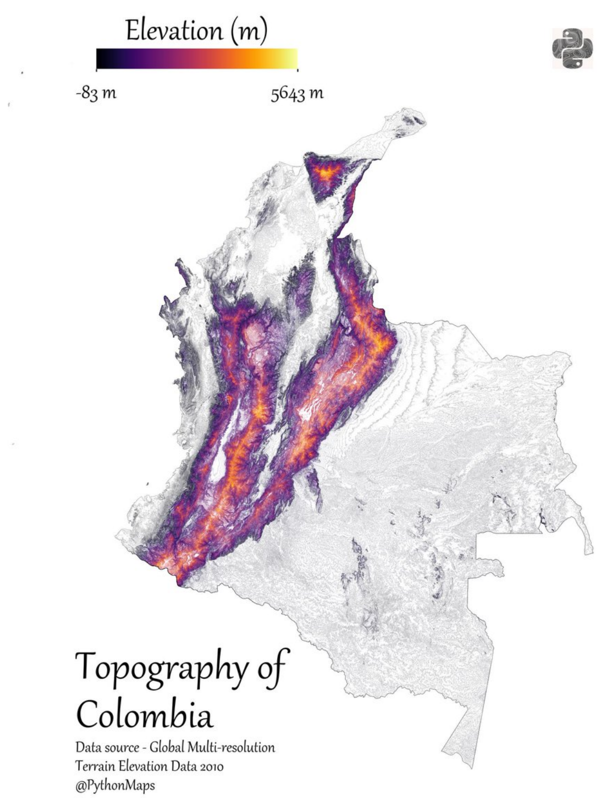
Virtually all of Eastern Colombia, about half of the country, is uninhabited road and city free pristine nature stretching from Eastern Arauca and Vichada down to Amazonas. Likewise, but much narrower, pristine nature belt stretches over much of Western Pacific, especially in Choco.

Colombian population is concentrated along 2 valleys in Andes–one along Panamericana highway from second largest Medellin

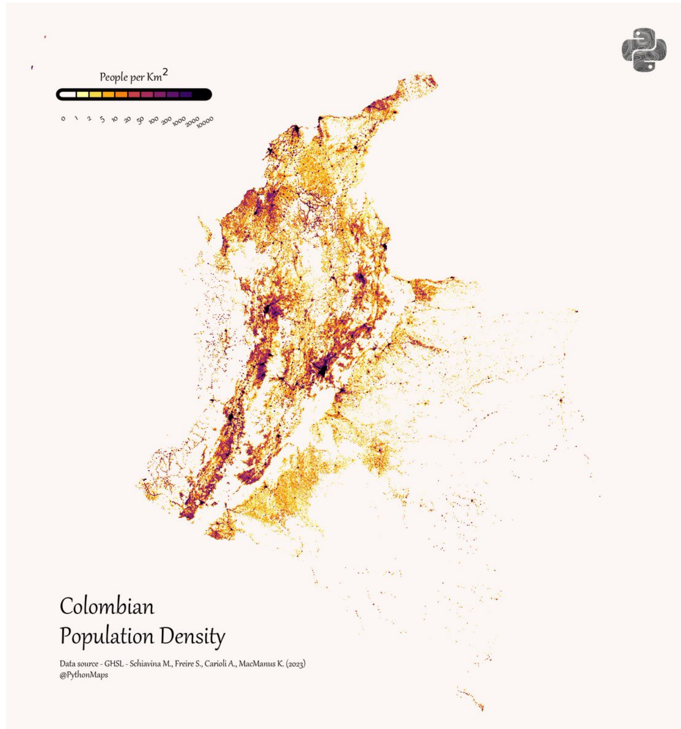
[8] 2018 quality of roads rank is 110/137 (`https://reports.weforum.org/pdf/gci-2017-2018-scorecard/WEF_GCI_2017_2018_Scorecard_EOSQ057.pdf`)–part of the problem is mountains.

Figure 14: Topography and population density. Both maps come from visually appealing PythonMaps: hi-res versions are at the below links. A 3d topography map is also at https://www.reddit.com/r/MapPorn/comments/u9gr9j/the_topography_of_colombia. And another topography map at https://upload.wikimedia.org/wikipedia/commons/3/3b/Colombia_Topography.png.

(a) Topography https://x.com/PythonMaps/status/1663937378782937088/photo/1



(b) Population density https://x.com/PythonMaps/status/1699089505133772803/photo/1



through Caffetero region, third largest Cali, down to Ipiales on border with Ecuador. The second valley runs from Tunja to largest Bogota down to Neiva. And there is also quite populated Cariabean.

Last but not least ecological footprint. https://data.footprintnetwork.org/ is a useful global overview, Colombian footprint per person is quite low at 2gha, 4x lower than that of the US at 8 gha, and it is stable since 1960s, but bio-capacity per person has declined from 11.4 gha in 1960 to 3.3 in 2024. https://happyplanetindex.org/countries/COL/ is another tool for cross-country comparisons.

Figure 15 shows useful hi-resolution visualizations of footprint across Colombia for 2015.[9]

[9] OECD provides more recent update across Colombia: https://www.oecd.org/en/publications/oecd-environmental-performance-reviews-colombia-2026_968398f7-en/full-report/biodiversity-conservation-and-sustainable-use_2099ac2c.html

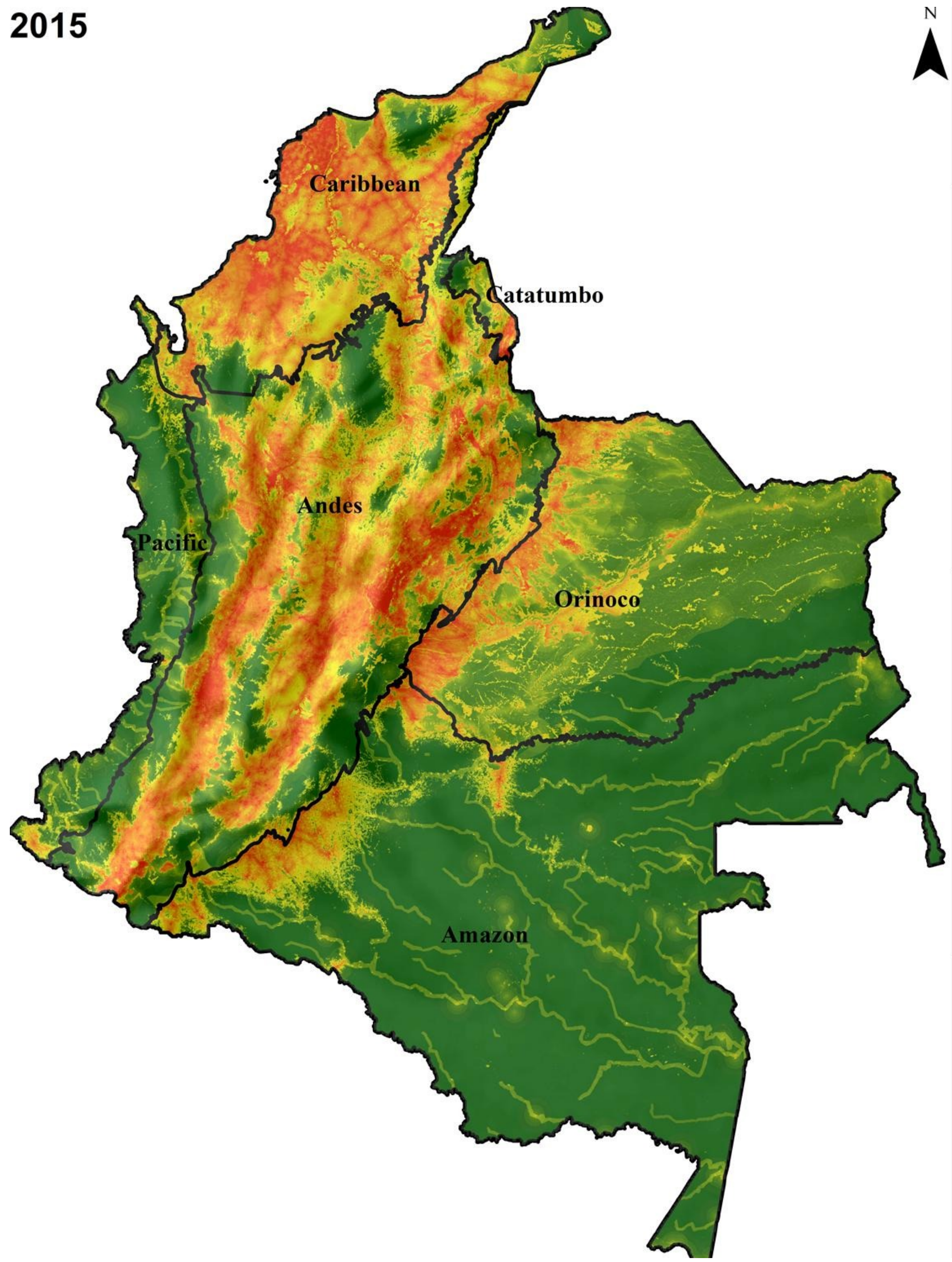
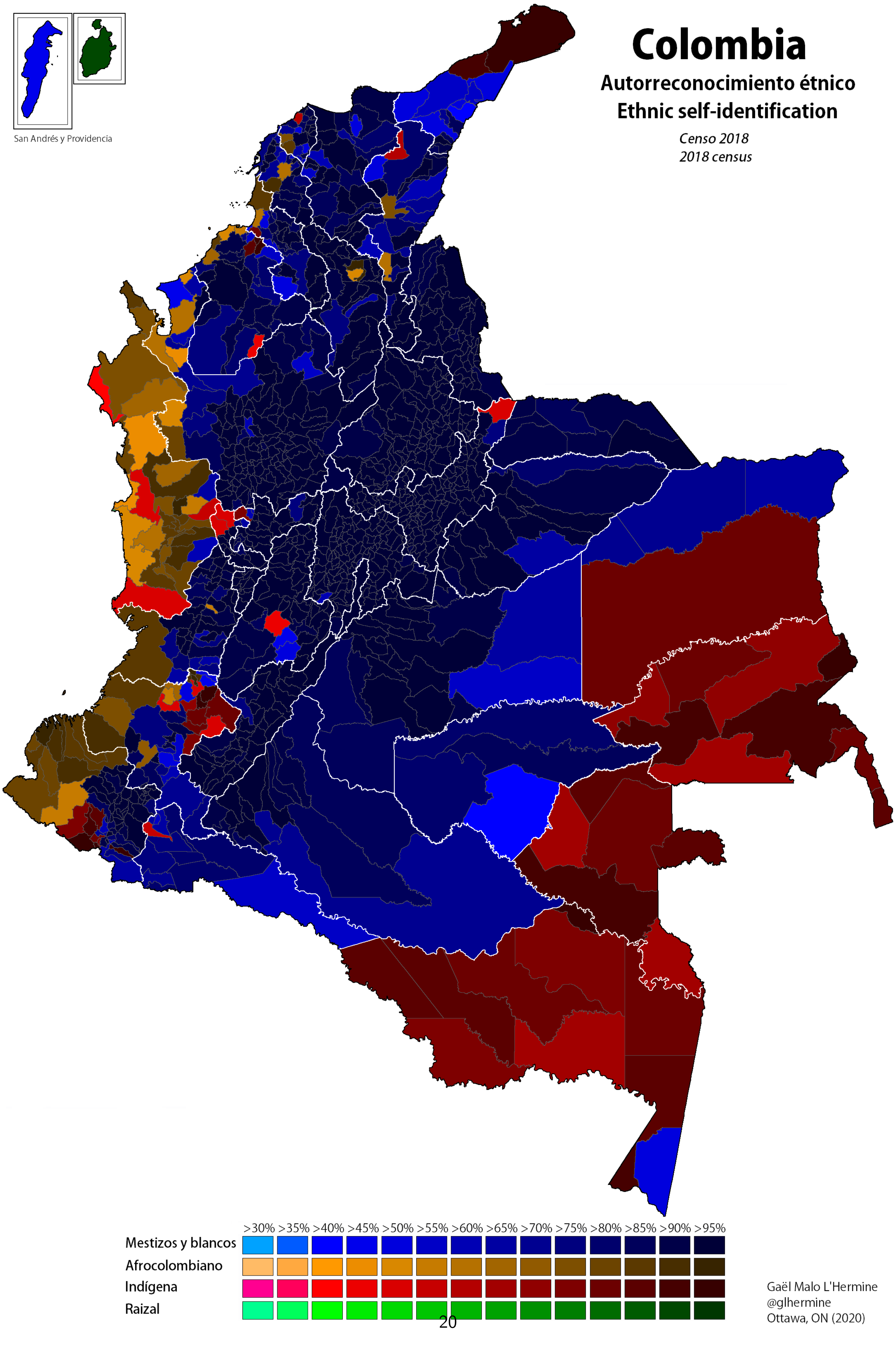


Figure 15: Human footprint: Low-Medium-High: Green-Yellow-Red. Source: `https://www.sciencedirect.com/science/article/pii/S1470160X20305677`

9.3 Socio-cultural Geography

Ethnic self-identification is shown in figure 16.



Colombia
Autorreconocimiento étnico
Ethnic self-identification
Censo 2018
2018 census
San Andrés y Providencia
>30% >35% >40% >45% >50% >55% >60% >65% >70% >75% >80% >85% >90% >95%
Mestizos y blancos
Afrocolombiano
Indígena
Raizal
Gaël Malo L'Hermine
@glhermine
Ottawa, ON (2020)

Colombia has extraordinary diversity, not just biodiversity (2nd largest in the world after 8x larger Brazil), but also cultural and social. Colombia can be thought of as a microcosm of Latin America culturally and socially[10] with Indigenous, African, and Spanish heritage. Indigenous heritage and culture is still alive and widespread, as opposed to the US where it was largely eliminated (Whittle 2026).

Administratively Colombia has 32 departments further divided into municipalities. They are very diverse, almost as if different countries. And there are large differences within departments and across their municipalities as well. For instance in Narino, Pasto is very different from Tumaco. Likewise, in Valle de Cauca, Cali is very different from Buenaventura.

very urban Bogota and moderately urban Santander are >20k, while very rural Amazonas and rural Choco are <7k–threefold difference.

Urbanization differs widely–and urbanization is not just built environment by way of life (Wirth 1938). Again about half of eastern Colombia is wilderness, then there are many small towns and villages with slow and communal way of life, and then there are also a handful of cities where way of life is more Western-like.

For instance, Bogota (or at least parts of it) is similar to other large fast paced commerce oriented cities like New York or London. Likewise, it is very international and has the most connected airport in Latin America (most international connections) and one of the 3 largest airports in Latin America (close to Sao Paulo and Mexico city).[11] Cartagena is Miami-like: expensive and fashionable catering to the rich and tourists. On the other hand, Cali, the third largest city, is very much Latin with more laid back atmosphere and salsa music. Indeed, one could think these cities belong to different countries or even cultures. Pacific Buenaventura and Tumaco are different yet–less organized and impoverished. One could enumerate many more examples of differences across municipalities.

Bogota could be considered a primate city, although the difference versus second largest Medellin is only of order of about 2, smaller than in many other countries[12]

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[10]See https://colombia.co/en/colombia-country/history/greatest-cultural-blend and Okulicz-Kozaryn and Martinez (2025). Also physically–the only South American country with both Pacific and Atlantic coasts, and has just about any type of topography, climate, and vegetation: Andes with several snowy peaks over 5km, hot and dry deserts, humid rain forests, and moderate climate valleys.

[11]https://www.oag.com/blog/latin-americas-most-connected-airports-aviation-infographics and https://en.wikipedia.org/wiki/List_of_the_busiest_airports_in_Latin_America.

[12]https://en.wikipedia.org/wiki/Primate_city.

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